

P8107 FINAL EXAM

Fall 2022

Name:

UNI:

For this exam you allowed up to four sheets of paper (front and back). No other reference materials may be used. You have 1 hour and 30 minutes to complete this exam. Be sure to show all your work. You may attach additional pages to your exam paper if needed.

Honor code:

I have not and will not give or receive aid in this examination nor have I concealed any violation of the University Honor Code.

Sign here: _____

1. (20 points) Random variables $Y_1, Y_2, \dots, Y_n$ have a pdf $f(y) = \frac{1}{\theta} y^{\frac{1-\theta}{\theta}}$. Find the maximum likelihood estimator of θ .

2. (10 points) A hypothesis testing procedure for $H_0: \mu = 18$ vs. $H_a: \mu > 18$ is to reject H_0 if $\bar{Y} > 19$. Calculate the power of this test for $n = 25$ and $\sigma^2 = 9$ if the true mean is 20. (For this problem it is okay to leave your answer in terms of the cdf of a standard normal distribution, e.g., $P(Z < 1.5)$.)

3. (20 points) To test $H_0: \mu_X - \mu_Y = 2$ vs. $H_a: \mu_X - \mu_Y \neq 2$ you will find that the likelihood ratio test is equivalent to rejecting H_0 if $\bar{X} - \bar{Y}$ is too “large” or too “small”, i.e., reject H_0 if $\bar{X} - \bar{Y} < K_1$ or if $\bar{X} - \bar{Y} > K_2$. Assuming that $X_1, X_2, \dots, X_m$ are iid $N(\mu_X, 1)$ and $Y_1, Y_2, \dots, Y_n$ are iid $N(\mu_Y, 1)$ (and that the X observations are independent of the Y observations), find K_1 and K_2 so that the probability of a Type I error is α .

4. (25 points) Random variables $Y_1, Y_2, \dots, Y_n$ are iid $N(\mu, \sigma^2)$ where both μ and σ^2 are unknown.
- The natural estimator of μ is $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$. Calculate the variance of $\bar{Y}$.
 - A colleague thinks that μ is probably about 10 and suggests an alternative estimator which consists of adding an extra “observation” that is equal to 10, i.e., $\hat{\mu}^* = \frac{1}{n+1} [(\sum_{i=1}^n Y_i) + 10]$. Your colleague claims that this estimator will have smaller variance than $\bar{Y}$. Calculate the variance of $\hat{\mu}^*$ and compare it with that of $\bar{Y}$. Is your colleague correct?
 - Calculate the MSE of both estimators.
 - Which estimator has better MSE if in fact $\mu = 10$?
 - Which estimator has better MSE if in fact $\mu = 9$ and $n = 4$?

5. (10 points) For observations $Y_1, Y_2, \dots, Y_n$ that are iid $N(\mu, \sigma^2)$, the maximum likelihood estimator of σ^2 is $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2$.
- Find $E[\hat{\sigma}^2]$ and $Var(\hat{\sigma}^2)$.
 - Based on (a), is $\hat{\sigma}^2$ a consistent estimator of σ^2 ? Explain why or why not.

6. (15 points) To estimate the probability p given a binomial random variable $Y \sim \text{binom}(n, p)$, the maximum likelihood estimator can be found to be $\hat{p} = \frac{Y}{n}$. Using the large-sample properties of MLEs, derive an approximate 95% confidence interval for p . (To receive credit for this problem, you must start with the expression for $\sigma_{\hat{\theta}}^2$ and derive the interval – not just write down the answer.)